

Univrsiy of Kalyani

Department of Computer Science and Engineering

Customer Lifetime Value Analysis using RFM Segmentation and Machine Learning

Project documentation submitted in partial fulfilment of the
requirements for the degree/coursework in

Computer Science and Engineering

Submitted by

Suman Banerjee

University Roll No.: **90/BCS/220060**

Registration No.: **1090001**

Under the supervision of

Asst. Prof. Sukanta Majumder

Department of Computer Science and Engineering

University of Kalyani

Kalyani, India

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Declaration

I, **Suman Banerjee**, University Roll No. **90/BCS/220060** and Registration No. **1090001**, declare that this project report entitled “**Customer Lifetime Value Analysis using RFM Segmentation and Machine Learning**” is a bona fide record of the work carried out by me under the supervision of **Asst. Prof. Sukanta Majumder**, Department of Computer Science and Engineering, University of Kalyani.

This report represents my own understanding, implementation, analysis, and interpretation of the project. Wherever external ideas, datasets, software libraries, or standard mathematical methods have been used, they have been acknowledged appropriately. I further declare that the work reported here has not been submitted, either in part or in full, for the award of any other degree or diploma.

Suman Banerjee
25.06.2026

Suman Banerjee

University Roll No.: 90/BCS/220060

Registration No.: 1090001

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Certificate

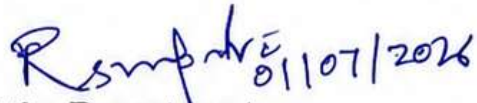
This is to certify that the project report entitled “**Customer Lifetime Value Analysis using RFM Segmentation and Machine Learning**” submitted by **Suman Banerjee**, Registration No. **1090001**, University Roll No. **90/BCS/220060**, in the Department of Computer Science and Engineering, University of Kalyani, is a bona fide work carried out under my supervision.

The project fulfils the expected academic requirements and, in my opinion, is suitable for presentation and evaluation.


Internal Supervisor

Asst. Prof. Sukanta Majumder

Department of CSE


Head of the Department
Professor & Head
Dept. of Computer Science & Engg.
University of Kalyani
Kalyani, Nadia-741 235
Department of Computer Science and
Engineering
University of Kalyani

External Examiner(s)




Abstract

Customer Lifetime Value (CLV) is an important business metric that estimates the total economic value a customer can generate during the period of relationship with a company. The objective of this project is to analyze an online retail transaction dataset, transform raw transactional data into customer-level business features, segment customers using Recency-Frequency-Monetary (RFM) analysis, estimate CLV, and predict CLV tiers using machine learning models.

The project uses the `online_retail_II.xlsx` dataset containing two Excel sheets for the periods 2009–2010 and 2010–2011. After cleaning missing customer identifiers, cancelled invoices, and invalid quantity or price values, the usable dataset contains **805,549** transactions from **5,878** customers. The project creates RFM scores, behavioral features, CLV values, and CLV tiers. Customers are grouped into business segments such as Champions, Loyal Customers, Potential Loyalists, Promising, At Risk, Churned Customers, and Regular Customers.

Machine learning models including Random Forest, Gradient Boosting, and XGBoost are trained to predict whether a customer belongs to Low, Medium, or High CLV tiers. The best model in the notebook is **Gradient Boosting**, achieving **97.70%** test accuracy. The final Streamlit dashboard presents data overview, Pareto analysis, RFM segmentation, CLV analysis, and machine learning evaluation through interactive visualizations.

Keywords: Customer Lifetime Value, RFM Analysis, Data Cleaning, Business Intelligence, Machine Learning, Gradient Boosting, XGBoost, Streamlit, Online Retail Analytics.

List of Abbreviations

Abbreviation	Meaning
CLV	Customer Lifetime Value
RFM	Recency, Frequency, Monetary
AOV	Average Order Value
ML	Machine Learning
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
GUI	Graphical User Interface
API	Application Programming Interface
XGBoost	Extreme Gradient Boosting

List of Symbols

Symbol	Description
R	Recency, measured as days since last purchase
F	Frequency, measured as number of unique invoices
M	Monetary value, measured as total customer revenue
CLV	Customer Lifetime Value
AOV	Average Order Value
n	Number of observations/customers
y	True target label
$\hat{y}$	Predicted target label
p	Predicted probability

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Chapter 1

Introduction

1.1 Project Overview

This project is based on customer analytics for an online retail business. The main purpose is to understand customer behavior from historical transaction data and convert that understanding into business actions. Instead of only reporting total sales, the project identifies which customers are valuable, which customers are likely to be retained, which customers may churn, and which customer groups should receive different marketing strategies.

The project has two major parts:

1. A Jupyter/Google Colab notebook, `CLV_NEW.ipynb`, used for data preparation, feature engineering, model training, evaluation, and artifact saving.
2. A Streamlit application, `app.py`, used for an interactive dashboard with business charts and model performance visualization.

1.2 Problem Statement

Retail businesses often have many customers, but all customers do not contribute equally to revenue. Some customers buy repeatedly and generate high revenue; some buy once and disappear; some were valuable earlier but have not purchased recently. A business needs a systematic method to identify these groups.

The problem solved in this project is:

To analyze online retail transaction data, segment customers using RFM behavior, calculate Customer Lifetime Value, and predict customer value tiers using machine learning.

1.3 Objectives

The main objectives of the project are:

- Load and combine multi-sheet Excel retail transaction data.
- Clean invalid records such as missing customer IDs, cancelled invoices, negative quantities, and invalid prices.
- Convert transaction-level data into customer-level analytical data.
- Compute RFM features: Recency, Frequency, and Monetary value.
- Create customer segments using RFM score rules.
- Estimate CLV using purchase behavior and customer lifespan.
- Create Low, Medium, and High CLV tiers.
- Train and compare machine learning classifiers for CLV tier prediction.
- Build an interactive Streamlit dashboard for business users.

1.4 Dataset Used

The dataset file used in the project is `online_retail_II.xlsx`. It contains two sheets:

- Year 2009–2010
- Year 2010–2011

The raw columns are `Invoice`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `Price`, `Customer ID`, and `Country`. In the code these are normalized to `InvoiceNo`, `UnitPrice`, and `CustomerID` for consistency.

Table 1.1: Dataset Summary

Metric	Value
Raw transaction rows	1,067,371
Cleaned transaction rows	805,549
Removed rows	261,822
Unique customers	5,878
Unique invoices	36,969
Unique products	4,631
Countries	41
Date range	01-Dec-2009 to 09-Dec-2011
Total revenue after cleaning	17,743,429.18

1.5 Why This Approach is Related to the Problem

The chosen approach is directly related to customer value analysis because CLV depends on how recently, how often, and how much a customer purchases. RFM analysis captures these three dimensions in an interpretable way. Machine learning then uses RFM and extra behavioral features to learn patterns that separate Low, Medium, and High CLV customers.

This combination is useful because:

- RFM gives explainable business segmentation.
- CLV converts customer behavior into monetary value.
- Pareto analysis identifies whether revenue is concentrated among a small group of customers.
- Machine learning predicts CLV tiers for future decision-making.
- Streamlit makes the analysis accessible through a dashboard.

Chapter 2

Literature Review and Core Concepts

2.1 Customer Lifetime Value

Customer Lifetime Value is a predictive business metric that estimates how much value a customer brings to a business over the relationship period. A simplified CLV formula used in this project is:

$$CLV = AOV \times F \times L \quad (2.1)$$

where AOV is Average Order Value, F is purchase frequency, and L is customer lifespan in years.

In the project:

$$AOV = \frac{M}{F} \quad (2.2)$$

Therefore:

$$CLV = \frac{M}{F} \times F \times L = M \times L \quad (2.3)$$

This means the implemented CLV is proportional to the total money spent by the customer and the observed duration of the customer relationship. A customer with high revenue and a long active purchasing period receives a higher CLV.

2.2 RFM Analysis

RFM analysis is a classical customer segmentation method:

- **Recency:** How many days have passed since the customer's last purchase.
- **Frequency:** How many unique invoices/purchases the customer made.
- **Monetary:** How much total revenue the customer generated.

The intuition is simple but powerful. Recent customers are more likely to respond to marketing, frequent customers show loyalty, and high monetary customers are

financially valuable.

2.3 Pareto Principle

The Pareto principle states that a large share of business results may come from a small share of causes. In retail analytics, it often appears as:

Approximately 80% of revenue may come from approximately 20% of customers.

The project tests this idea by sorting customers according to monetary value and calculating cumulative revenue. In the project output, the top 20% customers generated approximately **77.24%** of total revenue. This confirms strong revenue concentration.

2.4 Machine Learning Classification

The project treats CLV tier prediction as a supervised classification problem. Each customer is assigned a target label:

$$y \in \{Low, Medium, High\} \quad (2.4)$$

The machine learning model receives customer features X and learns a mapping:

$$f(X) \rightarrow y \quad (2.5)$$

After training, the model predicts the CLV tier for unseen customers.

2.5 Why Tree-Based Models Were Used

The project uses Random Forest, Gradient Boosting, and XGBoost. These models are suitable because customer behavior data is tabular, nonlinear, and may contain interactions. For example, high monetary value with low recency may mean a valuable current customer, while high monetary value with very high recency may indicate an at-risk customer.

Tree-based models handle such decision rules naturally. They also do not require strict feature scaling and can model nonlinear thresholds.

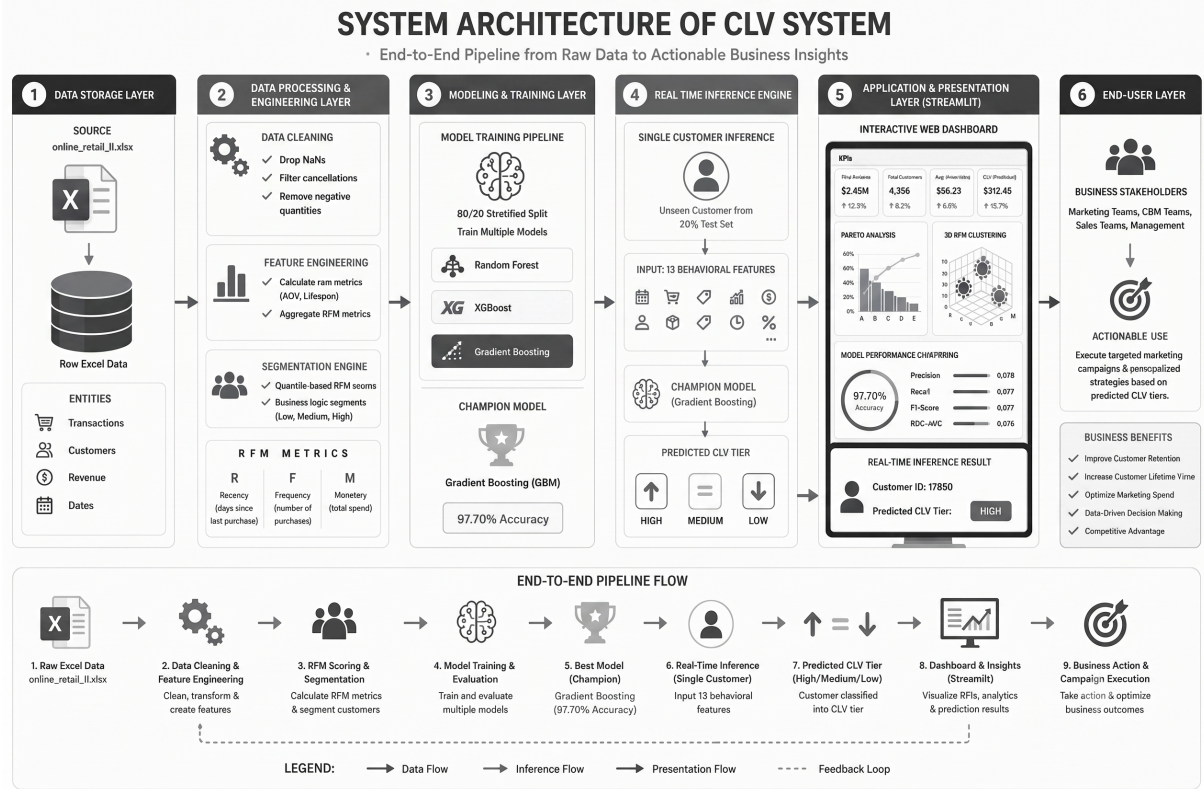
Chapter 3

Proposed Work and Methodology

3.1 System Architecture

The project workflow follows this pipeline:

1. Load Excel file containing two yearly sheets.
2. Concatenate sheets into one transaction table.
3. Rename columns for code consistency.
4. Clean missing and invalid transaction records.
5. Create `TotalPrice`.
6. Aggregate transaction data into customer-level RFM data.
7. Compute behavioral features and RFM scores.
8. Assign business segments.
9. Compute CLV and CLV tiers.
10. Train machine learning models.
11. Evaluate accuracy, classification report, confusion matrix, and feature importance.
12. Display results through a Streamlit dashboard.



3.2 Data Cleaning

Data cleaning is necessary because raw business data often contains returns, cancelled invoices, missing customer identifiers, and invalid values. The project performs the following cleaning:

Listing 3.1: Core data cleaning logic

```
df = df.dropna(subset=["CustomerID"])
df = df[~df["InvoiceNo"].astype(str).str.startswith("C")]
df = df[(df["Quantity"] > 0) & (df["UnitPrice"] > 0)]
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
df["CustomerID"] = df["CustomerID"].astype(int)
df["TotalPrice"] = df["Quantity"] * df["UnitPrice"]
```

3.2.1 Mathematical Meaning of Total Price

For each invoice line:

$$TotalPrice_i = Quantity_i \times UnitPrice_i \quad (3.1)$$

This converts product-level transaction rows into revenue values.

3.3 RFM Feature Engineering

The project defines a snapshot date as one day after the latest transaction date:

$$SnapshotDate = \max(InvoiceDate) + 1 \text{ day} \quad (3.2)$$

Recency for customer c is:

$$R_c = SnapshotDate - \max(InvoiceDate_c) \quad (3.3)$$

Frequency is:

$$F_c = \text{count of unique invoices for customer } c \quad (3.4)$$

Monetary value is:

$$M_c = \sum_{i \in c} TotalPrice_i \quad (3.5)$$

The code also creates additional behavior features:

- **Quantity_Total**: total items purchased.
- **Avg_UnitPrice**: mean item price.
- **Unique_Products**: product variety.
- **Invoice_Lines**: number of invoice lines.
- **Lifespan_Days**: observed relationship duration.
- **AOV**: average order value.
- **Avg_Basket_Size**: average quantity per invoice.
- **Purchase_Rate**: purchase frequency per active day.

3.4 RFM Scoring

The project uses quantile-based scoring with five groups. Quantiles divide customers into approximately equal-sized groups.

For Recency, lower values are better because a recent purchase indicates active engagement. Therefore the labels are reversed:

$$R_Score \in \{5, 4, 3, 2, 1\} \quad (3.6)$$

For Frequency and Monetary, higher values are better:

$$F_Score, M_Score \in \{1, 2, 3, 4, 5\} \quad (3.7)$$

The combined RFM score is a string:

$$RFM_Score = \text{concat}(R_Score, F_Score, M_Score) \quad (3.8)$$

A customer with score 555 is a very recent, very frequent, and very high monetary customer.

3.5 Customer Segmentation Rules

The project assigns business segments using RFM rules:

Table 3.1: Customer Segmentation Logic

Segment	Rule/Interpretation
Champions	RFM score is 555; best current customers.
Loyal Customers	High recency and high frequency.
Potential Loyalists	Recent customers who may become loyal.
Promising	Moderate recency and frequency.
Churned Customers	Low recency and low frequency.
At Risk	Old customers with poor recent activity.
Regular Customers	Remaining normal customers.

3.6 CLV Tier Creation

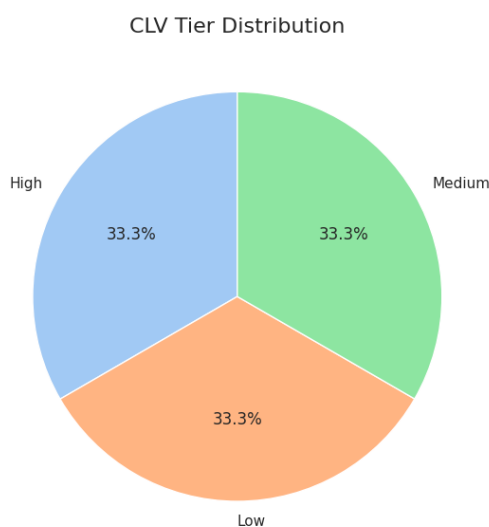
After calculating CLV, customers are divided into three quantile-based tiers:

$$CLV_Tier \in \{Low, Medium, High\} \quad (3.9)$$

Using quantiles keeps the target classes balanced. In this project, the tier distribution is nearly equal:

- Low: 1,960 customers

- Medium: 1,959 customers
- High: 1,959 customers



3.7 Machine Learning Feature Set

The model uses 13 features:

Feature	Meaning
Recency	Days since last purchase
Frequency	Number of unique invoices
Monetary	Total customer revenue
R_Score	Recency quantile score
F_Score	Frequency quantile score
M_Score	Monetary quantile score
Quantity_Total	Total units purchased
Avg_UnitPrice	Average product price purchased
Unique_Products	Number of different products bought
Invoice_Lines	Number of transaction rows
Avg_Basket_Size	Quantity per invoice
Avg_Lines_Per_Invoice	Invoice line density
Purchase_Rate	Purchases per active lifespan day

The notebook avoids using `Lifespan_Years` and `AOV` directly in the machine learning features because those columns are directly involved in the CLV formula. This reduces direct target leakage compared with training on the exact formula inputs.

3.8 Train-Test Split

The project uses an 80:20 train-test split:

$$Train = 80\%, \quad Test = 20\% \quad (3.10)$$

The actual split is:

- Training customers: 4,702
- Testing customers: 1,176

Stratification is used so that the class distribution remains balanced in both training and testing sets.

Chapter 4

Implementation and Code Documentation

4.1 Technologies Used

Table 4.1: Project Technology Stack

Technology	Purpose
Python	Main programming language
Pandas	Data loading, cleaning, aggregation, feature engineering
NumPy	Numerical operations
Scikit-learn	Machine learning models and evaluation metrics
XGBoost	Gradient boosting classifier
Plotly	Interactive dashboard charts
Matplotlib/Seaborn	Notebook visualizations
Streamlit	Web dashboard application
OpenPyXL	Reading Excel files
Joblib	Saving trained model artifacts

4.2 Requirements File

The `requirements.txt` file contains:

Listing 4.1: Project dependencies

```
streamlit >=1.30.0
pandas >=2.0.0
numpy>=1.24.0
scikit-learn >=1.3.0
xgboost >=2.0.0
plotly >=5.18.0
openpyxl >=3.1.0
```

4.3 Notebook Implementation

The notebook `CLV_NEW.ipynb` contains the experimental and training workflow. The major steps are:

1. Install/check XGBoost.
2. Import Python libraries.
3. Mount Google Drive in Colab.
4. Configure dataset path, random state, and test size.
5. Load both sheets from the Excel file.
6. Normalize column names.
7. Clean transactions.
8. Build RFM and behavior features.
9. Generate RFM scores and customer segments.
10. Visualize segment distribution.
11. Calculate CLV and CLV tiers.
12. Prepare machine learning features.
13. Train and compare classifiers.
14. Evaluate the best model.
15. Plot feature importance.
16. Save model and processed artifacts.
17. Predict CLV tier for a customer selected from the test set and verify whether the predicted tier correctly maps to the actual tier.

4.4 Streamlit Application Implementation

The file `app.py` converts the notebook analysis into an interactive dashboard. It uses the following cached functions:

- `load_and_clean_data(file_path)`: loads, combines, renames, and cleans the Excel data.
- `generate_rfm(df)`: creates customer-level RFM, behavior features, segments, CLV, and tiers.
- `train_models(rfm)`: trains Random Forest, Gradient Boosting, and XGBoost classifiers and returns evaluation outputs.

4.5 Dashboard Pages

The sidebar navigation contains five pages:

1. **Data Overview**: shows total transactions, customers, revenue, recent transactions, top countries, and revenue trend.
2. **Pareto Analysis**: calculates how much revenue is generated by the top 20% customers and displays a Pareto chart.
3. **RFM Segmentation**: shows segment distribution, segment statistics, and 3D RFM scatter plot.
4. **CLV Analysis**: shows CLV tier distribution, CLV spread, and average CLV by segment.
5. **ML Predictions**: trains models, compares accuracy, shows champion model, confusion matrix, and classification report.

4.6 Mathematical Intuition Behind Evaluation Metrics

4.6.1 Accuracy

Accuracy measures the proportion of correct predictions:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \quad (4.1)$$

The best model achieved:

$$Accuracy = 0.9770 = 97.70\% \quad (4.2)$$

4.6.2 Precision

Precision answers: out of all customers predicted as a class, how many were actually that class?

$$Precision = \frac{TP}{TP + FP} \quad (4.3)$$

4.6.3 Recall

Recall answers: out of all actual customers of a class, how many did the model correctly identify?

$$Recall = \frac{TP}{TP + FN} \quad (4.4)$$

4.6.4 F1-Score

F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4.5)$$

4.6.5 Confusion Matrix

A confusion matrix compares actual labels with predicted labels. For this project, the labels are High, Low, and Medium CLV tiers. It helps identify whether the model is confusing Medium customers with High or Low customers.

Chapter 5

Results, Business Insights, and Discussion

5.1 Data Cleaning Results

The raw dataset had 1,067,371 rows. After cleaning, 805,549 valid transaction rows remained. The removed records included missing customer identifiers, cancelled invoices, and non-positive quantity or price values.

Table 5.1: Cleaning Summary

Item	Count
Raw rows	1,067,371
Rows after cleaning	805,549
Rows removed	261,822
Rows with missing customer ID in raw data	243,007
Cancelled invoices in raw data	19,494
Rows with non-positive quantity or price in raw data	25,700

5.2 Revenue Insights

Table 5.2: Top Countries by Revenue

Country	Revenue
United Kingdom	14,723,147.52
EIRE	621,631.11
Netherlands	554,232.34
Germany	431,262.46
France	355,257.47
Australia	169,968.11
Spain	109,178.53
Switzerland	100,365.34
Sweden	91,549.72
Denmark	69,862.19

Business interpretation: The United Kingdom dominates revenue, which means the retail business is highly concentrated geographically. Marketing and inventory planning should focus strongly on the UK market, while countries such as EIRE, Netherlands, Germany, and France can be treated as promising international markets.

5.3 Pareto Insight

The top 20% customers account for **77.24%** of total revenue.

Table 5.3: Pareto Analysis

Metric	Value
Top 20% customer count	1,175
Revenue from top 20% customers	13,705,453.23
Share of total revenue	77.24%

Business interpretation: Revenue is strongly concentrated among a small high-value customer group. The business should protect this group through loyalty programs, early access offers, personalized communication, and priority service.

5.4 RFM Segment Results

Table 5.4: Customer Segment Distribution

Segment	Customers
Churned Customers	1,523
Loyal Customers	1,008
Potential Loyalists	882
At Risk	824
Promising	782
Champions	474
Regular Customers	385

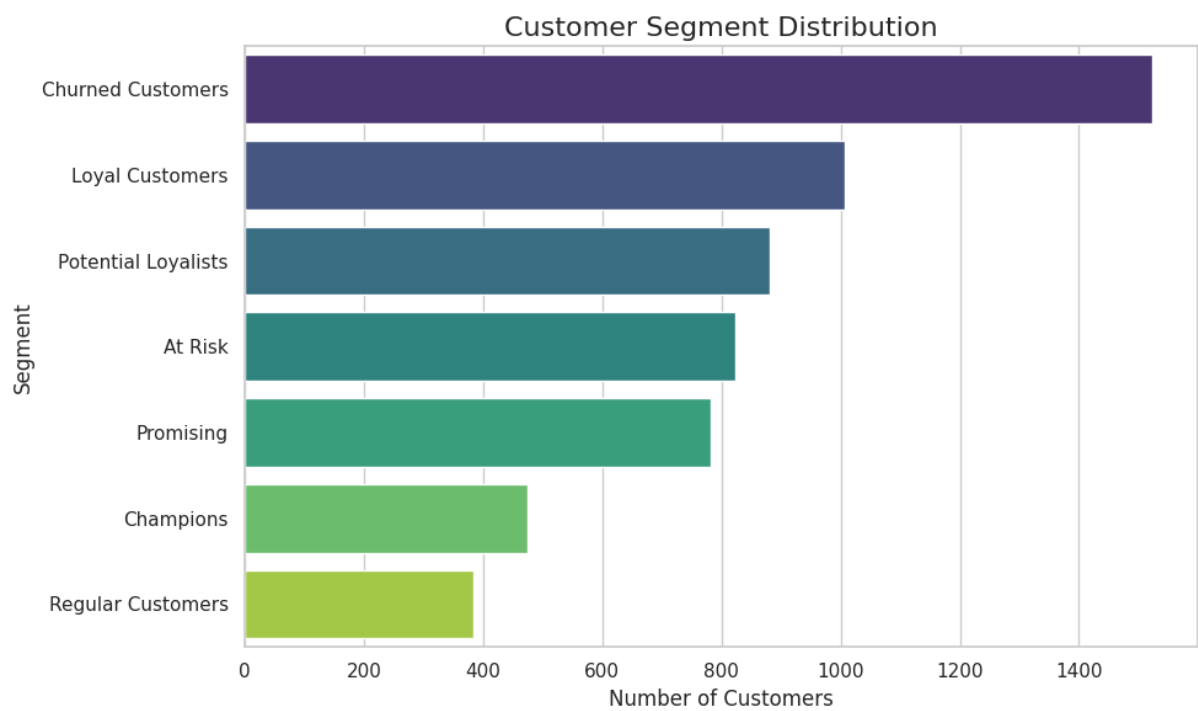


Table 5.5: Segment-Level Business Statistics

Segment	Count	Avg. Recency	Avg. Freq.	Avg. Monetary	Avg. CLV
Champions	474	7.85	28.56	17,630.55	32,631.42
Loyal Customers	1,008	26.25	9.59	3,905.02	6,081.74
Promising	782	108.70	6.70	2,637.47	3,619.01
At Risk	824	369.10	4.94	1,982.98	1,552.28
Potential Loyalists	882	26.51	2.29	998.95	625.18
Regular Customers	385	107.11	1.36	534.07	187.90
Churned Customers	1,523	459.28	1.25	438.03	61.46

5.5 Segment-Wise Business Actions

Segment	Meaning	Recommended Business Action
Champions	Most recent, frequent, and valuable customers.	Give premium loyalty benefits, exclusive offers, and referral rewards.

Loyal Customers	Frequent and recent buyers.	Encourage upselling, bundles, and membership programs.
Potential Loyalists	Recent but not yet very frequent.	Send onboarding offers and product recommendations.
Promising	Moderate activity and potential.	Nurture with targeted campaigns and reminders.
At Risk	Previously valuable but inactive recently.	Use win-back campaigns, discounts, and feedback surveys.
Churned Customers	Low recency and low frequency.	Use low-cost reactivation campaigns; avoid overspending.
Regular Customers	Ordinary customers with limited value.	Maintain periodic engagement and broad promotions.

5.6 CLV Distribution

The customer base is divided into three balanced CLV tiers:

Table 5.7: CLV Tier Distribution

CLV Tier	Customers
Low	1,960
Medium	1,959
High	1,959

The descriptive statistics show that CLV is highly skewed. The mean CLV is 4,495.44, while the median CLV is 496.49. This difference indicates that a small number of very high-value customers pull the average upward.

Table 5.8: Selected RFM and CLV Descriptive Statistics

Metric	Mean	Median	75th Percentile	Maximum
Recency	201.33	96.00	380.00	739.00
Frequency	6.29	3.00	7.00	398.00
Monetary	3,018.62	898.92	2,307.09	608,821.65
CLV	4,495.44	496.49	2,670.22	1,232,655.34
AOV	391.73	285.07	420.57	84,236.25

5.7 Machine Learning Results

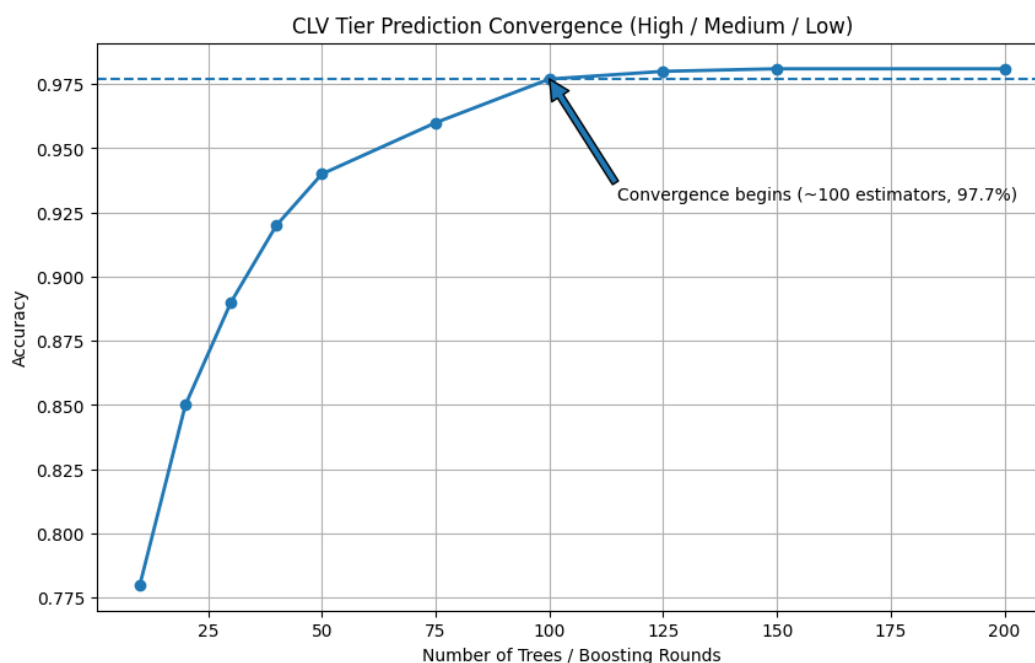
Table 5.9: Model Accuracy Comparison

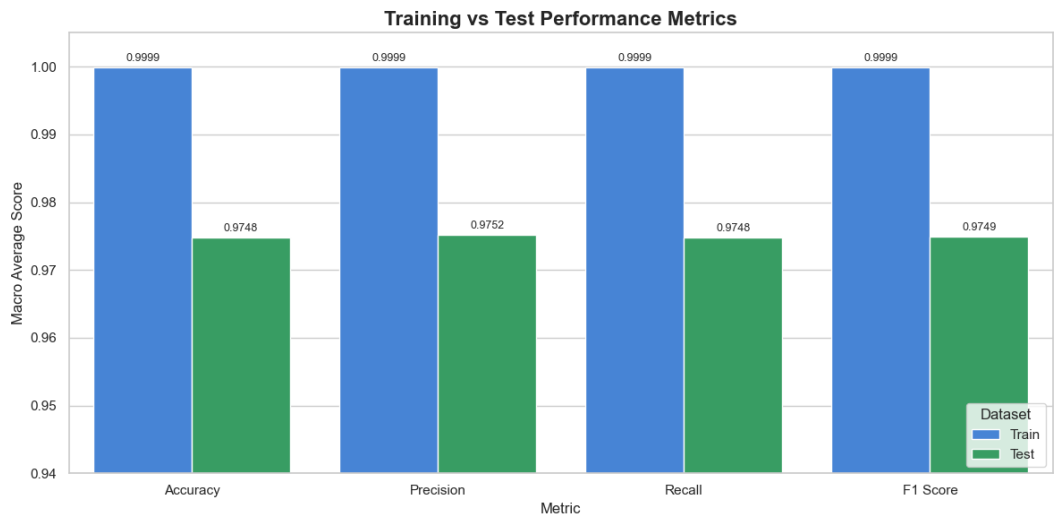
Model	Accuracy
Gradient Boosting	0.977041
XGBoost	0.977041
Random Forest	0.964286

The best selected model in the notebook is **Gradient Boosting**, with a test accuracy of **97.70%**.

Table 5.10: Classification Report for Best Model

Class	Precision	Recall	F1-score	Support
High	0.9820	0.9719	0.9769	392
Low	1.0000	0.9770	0.9884	392
Medium	0.9506	0.9821	0.9661	392
Accuracy			0.9770	1,176
Macro Avg	0.9775	0.9770	0.9771	1,176
Weighted Avg	0.9775	0.9770	0.9771	1,176





Only Gradient Boosting Results

Table 5.11: Confusion Matrix

	Predicted High	Predicted Low	Predicted Medium
Actual High	381	0	11
Actual Low	0	383	9
Actual Medium	7	0	385

5.8 Feature Importance

Table 5.12: Best Model Feature Importance

Feature	Importance
Purchase_Rate	0.486594
Monetary	0.383886
Frequency	0.093292
M_Score	0.015017
Recency	0.012955
F_Score	0.003978
Quantity_Total	0.001533
Avg_Basket_Size	0.001064
Invoice_Lines	0.000688
Unique_Products	0.000526
Avg_UnitPrice	0.000319
Avg_Lines_Per_Invoice	0.000136
R_Score	0.000012

Interpretation: Purchase rate and monetary value are the most important predictors. This is logical because CLV depends strongly on how much money a customer generates and how actively the customer purchases during the observed relationship.

5.9 Final Test-Set Prediction Verification

The last step of the notebook predicts the CLV tier for one customer taken from the test set. This is an important project submission point because it shows that the trained model is not only evaluated by accuracy, but is also used on an individual customer record and mapped back to a meaningful business tier: High, Medium, or Low.

For the reproduced notebook run, the selected customer from `X_test.iloc[0]` is **CustomerID 15311**. The customer's actual CLV tier is **High**, and the model prediction is also **High**. Therefore, the final prediction is correctly mapped.

Table 5.13: Individual Test-Set Customer Prediction

Item	Value
CustomerID	15311
Actual CLV Tier	High
Predicted CLV Tier	High
Mapping Status	Correct
Customer CLV	236,421.61
Recency	1 day
Frequency	208 invoices
Monetary Value	116,771.16
RFM Score	555

Business interpretation: CustomerID 15311 is a very high-value customer because the customer purchased recently, purchased frequently, generated high revenue, and received an RFM score of 555. The model correctly identifies this customer as **High CLV**, meaning the customer should be prioritized for loyalty benefits, retention campaigns, premium service, and personalized offers.

5.10 Overall Business Insights

The main business insights from the project are:

- The customer base is revenue-skewed; a small group of customers contributes most of the revenue.
- Champions are the most valuable customers, with average monetary value of 17,630.55 and average CLV of 32,631.42.

-
- Churned Customers are the largest segment, with 1,523 customers, but their average CLV is only 61.46.
 - At Risk customers have meaningful historical value but high recency, so they should be targeted before they fully churn.
 - Potential Loyalists are recent customers with low frequency, so they are good candidates for retention and repeat-purchase campaigns.
 - The best model can classify CLV tiers with high accuracy, which allows the business to automate customer prioritization.

Conclusion and Future Scope

6.1 Conclusion

This project successfully transforms raw online retail transaction data into business-ready customer intelligence. The work begins with data cleaning and moves through RFM feature engineering, customer segmentation, CLV calculation, machine learning prediction, and dashboard deployment.

The analysis shows that customer value is highly concentrated. The top 20% customers generate approximately 77.24% of total revenue. RFM segmentation identifies clear business groups such as Champions, Loyal Customers, At Risk customers, and Churned Customers.

The machine learning model predicts CLV tiers with 97.70% accuracy, showing that the engineered features are highly informative.

The Streamlit dashboard makes the project practical because it allows users to explore the dataset, view customer segments, analyze CLV distribution, and compare model performance interactively.

6.2 Limitations

The main limitations are:

- The CLV formula is based on historical observed behavior and does not include profit margin, acquisition cost, or discount rate.
- The model predicts CLV tiers created from the same historical dataset; future real-world validation would be needed.
- Customer demographics, marketing campaign history, and web behavior are not available in the dataset.
- The dashboard retrains models during app execution, which may be slow on low-resource systems.

6.3 Future Scope

Future improvements can include:

- Adding profit margin to estimate profit-based CLV.
- Building time-based validation to test future prediction quality.
- Using probability outputs to rank customers by risk and opportunity.
- Adding customer churn prediction as a separate target.
- Deploying the Streamlit app on Streamlit Community Cloud or another hosting platform.
- Using sequence models or deep learning if timestamped customer journeys become available.
- Adding SHAP explainability for more detailed model interpretation.
- Saving trained models from the notebook and loading them in the dashboard instead of retraining every time.

How to Compile and Run the Project

A.1 LaTeX Compiler for This Documentation

This documentation is written in LaTeX. It can be compiled using:

- **Overleaf**: recommended for easiest compilation.
- **MiKTeX + TeXworks** on Windows.
- **TeX Live** on Windows/Linux/macOS.
- **pdfLaTeX compiler**: recommended compiler for this file.

The required logo file is:

`kalyani_logo.png`

Keep `CLV_Project_Documentation.tex` and `kalyani_logo.png` in the same folder before compiling.

A.1.1 Compile Command

From the folder containing this file, run:

Listing A.1: Compile LaTeX document

```
pdflatex CLV_Project_Documentation.tex  
pdflatex CLV_Project_Documentation.tex
```

The second run updates the table of contents correctly.

A.2 How to Run the Notebook

The notebook is designed for Google Colab. Upload `online_retail_II.xlsx` to:

`/content/drive/MyDrive/CLV_Project/online_retail_II.xlsx`

Then run the notebook cells from top to bottom.

A.3 How to Run the Streamlit Dashboard

Install dependencies:

Listing A.2: Install dependencies

```
pip install -r requirements.txt
```

Run the dashboard from the project folder:

Listing A.3: Run Streamlit app

```
streamlit run app.py
```

The file `online_retail_II.xlsx` should be present in the same folder as `app.py`, because the app uses:

```
data_path = "online_retail_II.xlsx"
```

Appendix

B

Important

Source

Codes

B.1 RFM Generation

Listing B.1: RFM customer table creation

```
snapshot_date = df["InvoiceDate"].max() + pd.Timedelta(days=1)

rfm = (
    df.groupby("CustomerID")
    .agg(
        Recency=("InvoiceDate", lambda x: (snapshot_date - x.max
            ()).days),
        Frequency=("InvoiceNo", "nunique"),
        Monetary=("TotalPrice", "sum"),
        First_Purchase=("InvoiceDate", "min"),
        Last_Purchase=("InvoiceDate", "max"),
        Quantity_Total=("Quantity", "sum"),
        Avg_UnitPrice=("UnitPrice", "mean"),
        Unique_Products=("StockCode", "nunique"),
        Invoice_Lines=("InvoiceNo", "count"),
        Country=("Country", lambda x: x.mode().iloc[0] if not x.
            mode().empty else x.iloc[0]),
    )
    .reset_index()
)
```

B.2 CLV Calculation

Listing B.2: CLV and tier creation

```
rfm["Lifespan_Days"] = (rfm["Last_Purchase"] - rfm["
    First_Purchase"]).dt.days + 1
rfm["Lifespan_Years"] = rfm["Lifespan_Days"] / 365.0
rfm["AOV"] = rfm["Monetary"] / rfm["Frequency"]
```

```
rfm["CLV"] = rfm["AOV"] * rfm["Frequency"] * rfm["Lifespan_Years"]
rfm["CLV_Tier"] = pd.qcut(rfm["CLV"], q=3, labels=["Low", "Medium", "High"])
```

B.3 Model Training

Listing B.3: Model comparison approach

```
models = {
    "Random Forest": RandomForestClassifier(...),
    "Gradient Boosting": GradientBoostingClassifier(...),
    "XGBoost": XGBClassifier(...)
}

for model_name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
```

B.4 Final Prediction Function

Listing B.4: Prediction of one customer from the test set

```
def predict_clv_tier(customer_features):
    """Predict CLV tier for one customer.

    customer_features must contain all columns listed in
    feature_columns.
    """
    one_row = pd.DataFrame([customer_features])[feature_columns]
    prediction_id = best_model.predict(one_row)[0]
    return label_encoder.inverse_transform([prediction_id])[0]

example_customer = X_test.iloc[0].to_dict()
example_prediction = predict_clv_tier(example_customer)

print("\nExample prediction:")
```

```
print(example_prediction)
```

This final function proves the practical prediction flow. The model first predicts the encoded class ID, then `label_encoder.inverse_transform()` converts that numeric class back to the original CLV tier name. In the test-set example, the model predicted **High**, and the actual tier was also **High**, so the mapping was correct.

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